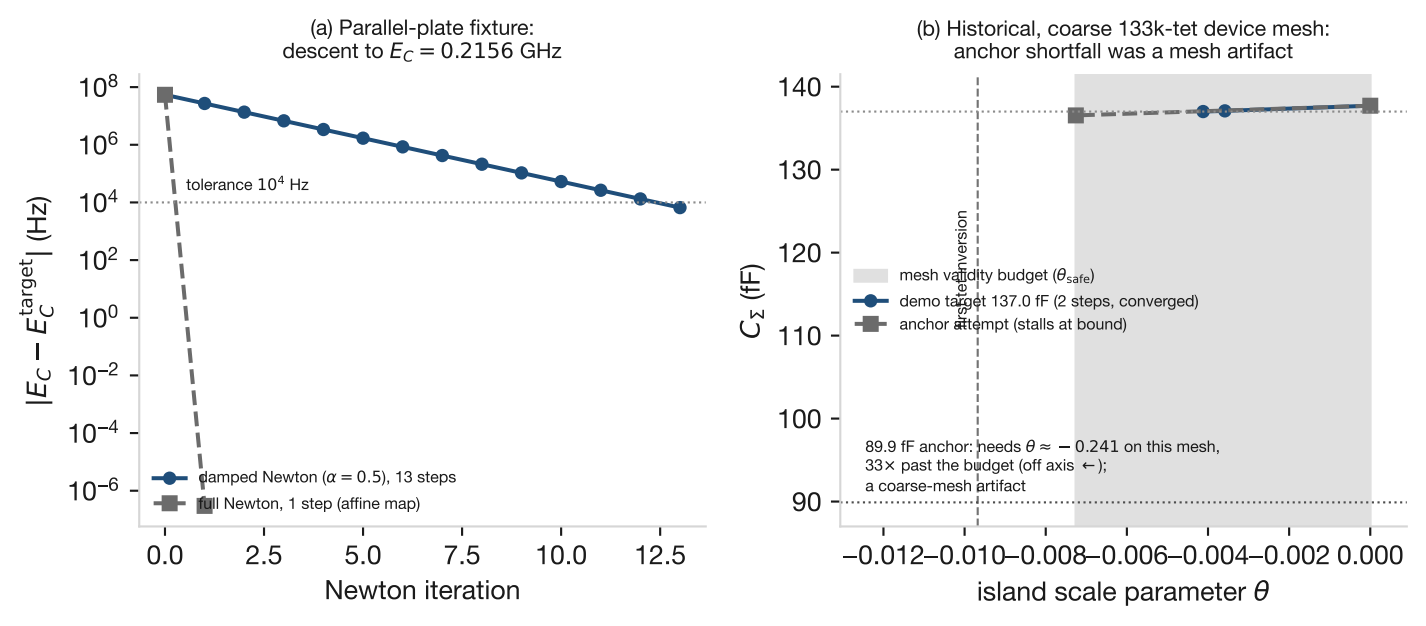
(a) Parallel-plate fixture:
descent to $E_C = 0.2156$ GHz
$|E_C - E_C^{\text{target}}|$ (Hz)
10^{8}
10^{6}
10^{4}
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
tolerance 10^{4} Hz
damped Newton ($\alpha = 0.5$), 13 steps
full Newton, 1 step (affine map)
0.0
2.5
5.0
7.5
10.0
12.5
Newton iteration
(b) Historical, coarse 133k-tet device mesh:
anchor shortfall was a mesh artifact
C_Σ (fF)
140
130
120
110
100
90
first inversion
mesh validity budget (θ_{safe})
demo target 137.0 fF (2 steps, converged)
anchor attempt (stalls at bound)
89.9 fF anchor: needs $\theta \approx -0.241$ on this mesh,
33× past the budget (off axis ←);
a coarse-mesh artifact
−0.012
−0.010
−0.008
−0.006
−0.004
−0.002
0.000
island scale parameter θ